# Malaria Detection

FINAL REPORT

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# Introduction

## Malaria detection problem statement - Context

Malaria is a life-threatening disease caused by Plasmodium parasites. Malaria is not contagious; it is transmitted to humans through the bites of female Anopheles mosquitoes. The risk of infection is higher in some locations due to the species of local mosquitoes, and during the rainy season in tropical countries.

Malaria is preventable and curable. Early diagnosis and treatment can be effective in improving outcomes for infected patients. Without early and effective treatment, malaria can be fatal. In the latest WHO World Malaria Report, there were 247 million cases in 2021 and 619,000 deaths. The WHO African Region has about 96% of all deaths and about 80% of the malaria deaths in the Region were children under the age of 5 years.

# Introduction

## Malaria detection problem statement – Diagnostic testing

There are two types of tests for the presence of the Plasmodium parasites. Rapid Diagnostic Tests (RDTs) are in-vitro tests that detect antigens from the parasites in patient blood samples.

Light microscopy allows healthcare providers to visually identify malaria-causing parasites. A patient blood sample from a finger prick is applied to a microscope slide and stained to improve human visualization of the parasite. This method is the gold-standard for malaria diagnosis. However, automation may aid in the detection and classification of the parasites by improving accuracy and reducing variability of humans.

## Malaria detection problem statement – Objective

*The objective of this project is to develop an efficient computer vision model to detect malaria and classify red blood cell images as parasitized or uninfected.*

# Data Report

- The images are labeled and come from two classes: parasitized and uninfected.
- We have 12,479 images from each class for training and 1300 images from each class for testing.
- Images are saved in png (Portable Network Graphic) format and are uint8 data type and have RGB (Red, Green, Blue) color space. The images, with 8 bits/color channel, can represent over 16 million colors (256**3).
- The image sizes vary. The x dimensions range from 46 to 394 pixels. The y dimensions range from 40 to 385 pixels. This means if a deep learning method like CNN is used, the images will need to be resized.
- Images appear to have different ranges of values. This suggests scaling will be needed in preprocessing images.
- *Some Things We Do Not Know*: Are all the images from different patients? In what part of the world were the patient samples collected? Were the images collected using the same lab and microscopy techniques? Are the samples stained? What color is the stain? How does parasite uptake of stain affect image?

# Initial Exploratory Data Analysis

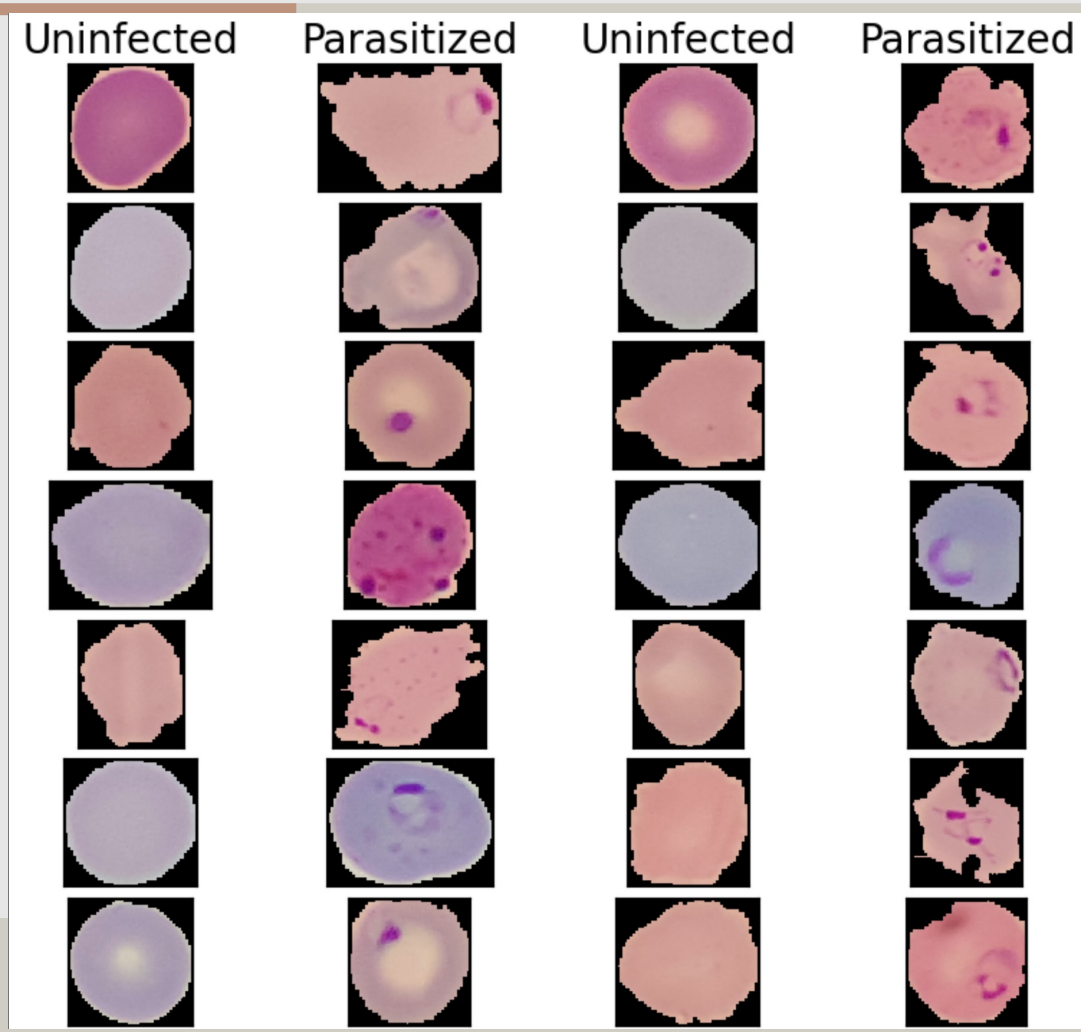
## Exploratory Methods Used

- Metadata were collected for all the images. This allowed summary descriptions of the file types, color spaces and x and y dimensions of the images. These numbers are reported in the Data Report section of this report.
- Visualization of raw images. This allowed examination of a large number of original images. From this we could learn more about visual features within class and between classes.
- Visualization of images before and after contrast enhancement using histogram equalization. This allowed us to see if one image processing technique could improve the visual features within class and between classes.
- Color and grayscale histograms. These allowed us to learn something about how the colors and grayscales are distributed within class and between classes.

# Initial Exploratory Data Analysis

## Image Visualization – Examples

- What characteristics are similar within each class and different between classes?
- Could traditional image processing and feature extraction work well with machine learning classifiers or is deep learning the best approach? Which approach would be faster? More computationally and memory efficient?



# Initial Exploratory Data Analysis

## Image Visualization - Observations

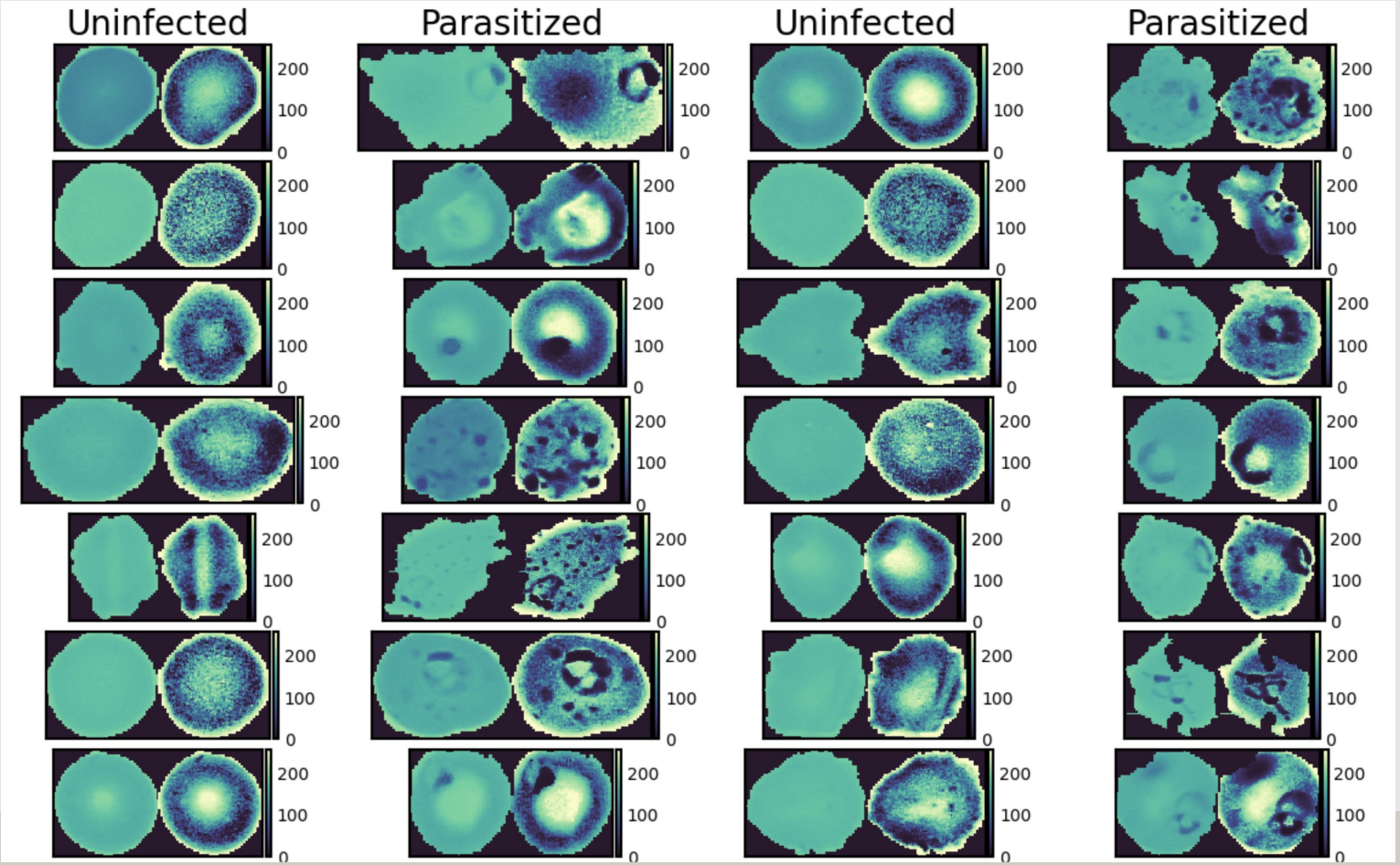
- The images all have black areas around the edges. They appear to be regions of interest (ROIs) extracted from other original images.
- Many diseased images appear to have spots - presumably the parasites. These may appear as distinct modes in the image histograms. This suggests a feature extraction method with ML classifier, perhaps a neural network, might work. This approach might be faster than a large deep learning network such as CNN.
- In these diseased images the spots occur in different locations in the image; not all in the center, for example. This suggests spatially invariant methods will be needed.
- Not all of the parasitized images have readily visible spots. In these cases, they appear to have a very uniform color across the cell and may be confused with uninfected cells leading false negatives in the classifier.

# Initial Exploratory Data Analysis

## Image Visualization with Contrast Enhancement

- Could traditional image processing and such as histogram equalization improve performance of the classifier?
- What about other enhancements? Pre-whitening? Other filters?

These images show the uninfected and parasitized images before and after histogram equalization side by side.



# Initial Exploratory Data Analysis

## Image Visualization with Contrast Enhancement – Observations

- These images show the uninfected and parasitized images before and after histogram equalization side by side.
- The method enhances the contrast in the images. In this case, the parasites are more distinct in the parasitized images which may improve recall. However, background is enhanced in the uninfected images which may cause false positives and reduce precision performance.

# Data Preprocessing

## Training, Validation and Test Datasets

- Training dataset comprises 24,958 images belonging to 2 classes
  - 18,719 (75%) images for training: 9,313 Parasitized and 9,406 Uninfected
  - 6,239 (25%) images reserved for validation: 3,166 Parasitized and 3,073 Uninfected
- Test dataset comprises 2,600 images belonging to 2 classes
  - 1,300 Parasitized and 1,300 Uninfected

### *Resizing images*

- All images are resized to 224 x 224 pixels in the datasets loaded from the directories

### *Normalizing images*

- All images are normalized to [0,1] by a keras.layers.Rescaling() layer in the CNN design

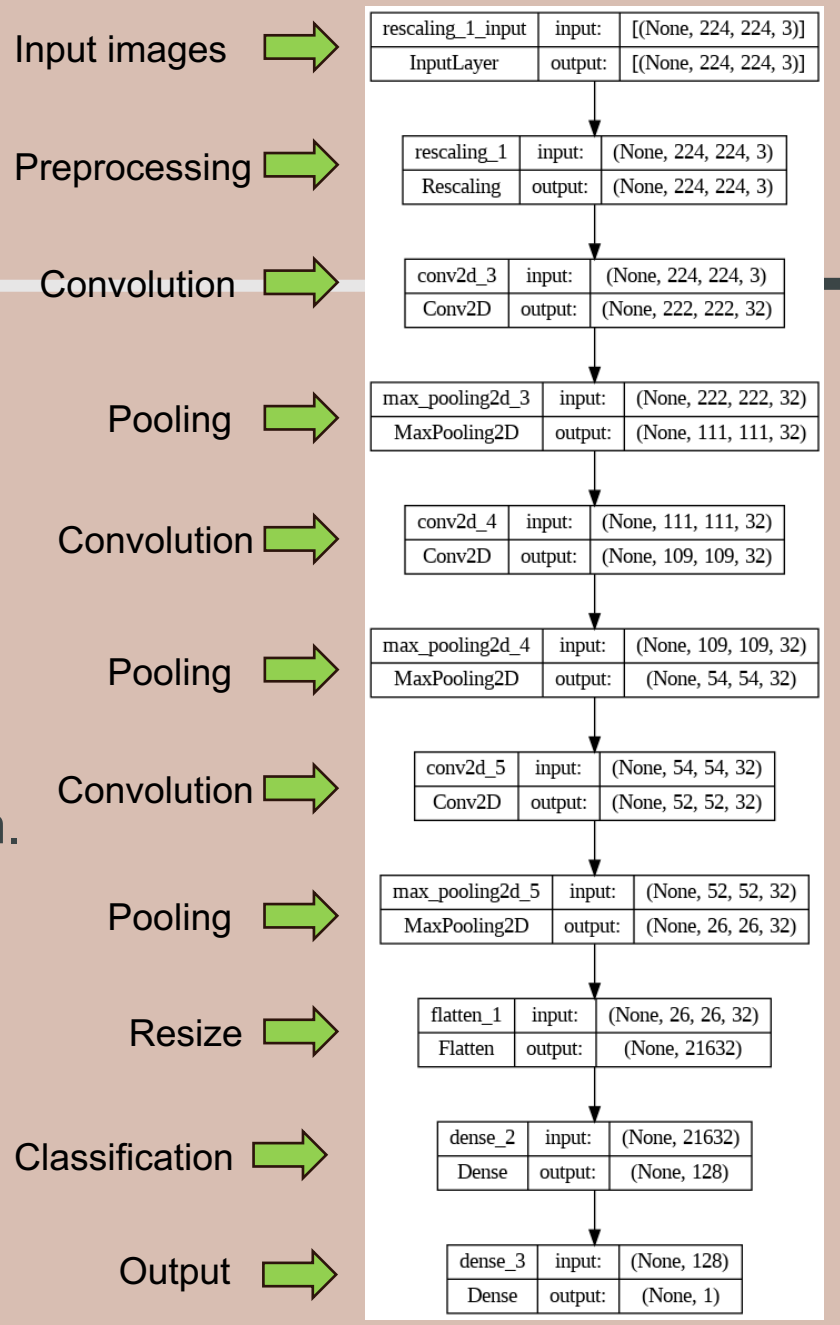
### *Encoding the class variable*

- The class variable for Parasitized images is 1.0 and for Uninfected images is 0.0

# Baseline Model

## Baseline CNN Model Architecture and Compiling

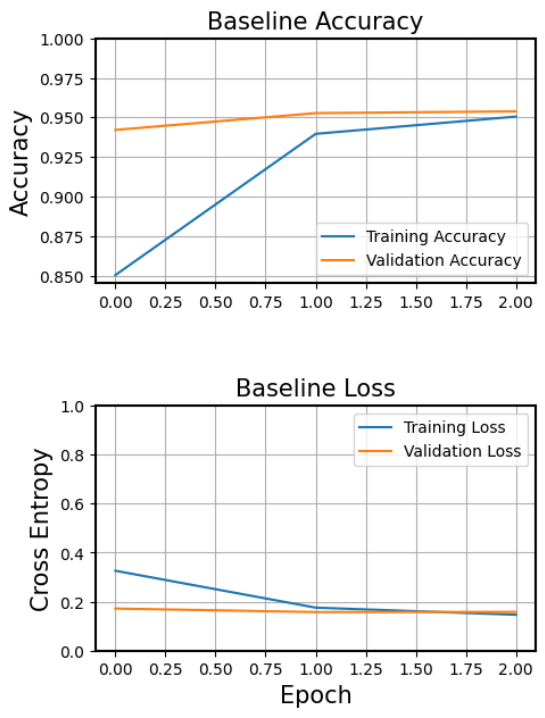
- The baseline CNN model provides a basis for comparing performance effects of techniques applied in the Exploratory Data Analysis.
- The model has a Rescaling input layer followed by 3 Convolutional / Max Pooling blocks for feature extraction and topped with a 128-node Dense layer for classification.
- Optimizer: Adam
- Loss function: BinaryCrossEntropy
- Initial metrics: BinaryAccuracy



# Baseline Model

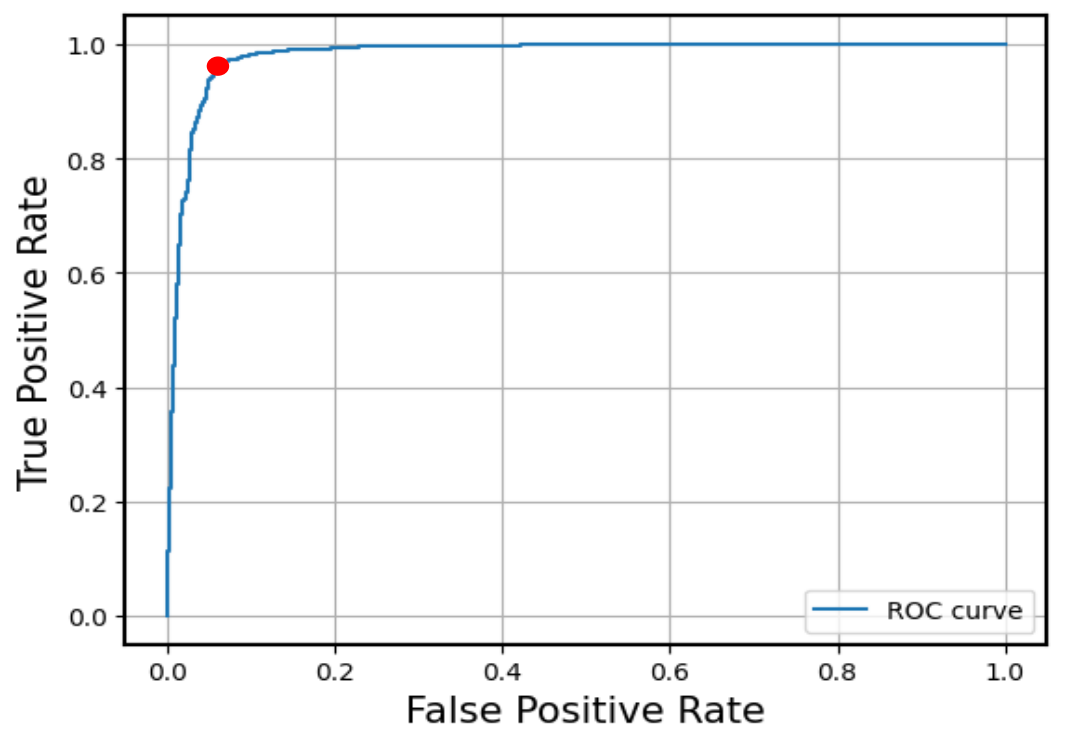
## Training

- Initial training for 4 epochs indicated some overfitting after epoch 3.
- The model was re-trained for 3 epochs and saved to disk for later comparisons.




## Testing

- The binary_accuracy with the test dataset is 0.945.
- The Area Under the ROC Curve (AUC) is 0.98



# Analytical Approach & Modeling Process

Use the Baseline Model to evaluate the effects of techniques aimed at improving performance

- *Image enhancement preprocessing methods are explored for improvement in baseline performance.*

Exploring Transfer Learning models

- *Transfer learning using a pre-trained models and modified architectures are explored.*

Candidate models with the best performance are saved and their metrics tabulated

- *Identify optimal performance threshold on the ROC curve. Metrics include indicators of the ROC curve 'shape' to help find the best tradeoff between TPR and FPR.*

Final Model selection

- *Best recall with the least increase in false positives wins.*

# Modeling Process

*Exploring potential improvements to the baseline model accuracy*

- Image processing enhancements:
  - Contrast enhancement
  - Gamma correction
  - Gamma correction followed by contrast enhancement
- Architecture: Additional Convolutional blocks, Increasing output filters by block, Dropout layers
- Activation functions: ReLU, Leaky ReLU, elu
- Callbacks: Adaptive learning schedule, Early stopping
- Increasing the number of training epochs
- Optimizers: Adam, Adagrad, RMSprop

These adjustments were explored in a large variety of combinations. The resulting models with the best performance were saved and the metrics tabulated.

# Best Performers

Best Result

| Model | Baseline | Contrast Enhancement with Baseline | Gamma and Contrast Enhancement with Baseline | Gamma Correction with Baseline using Leaky ReLU Activation | Gamma Correction with Baseline | Gamma Correction with Baseline Rev 1 | Gamma Correction with Baseline Rev 2 |
|---|---|---|---|---|---|---|---|
| Description | Baseline CNN trained for 3 Epochs | Preprocessing with contrast enhancement. Baseline architecture. | Processing with Gamma and Contrast Enhancement. Baseline architecture | Processing with Gamma correction. Baseline using Leaky ReLU | Processing with Gamma correction. Baseline architecture. | Modified architecture with increasing numbers of output filters in Conv2D layers. | Added a Dropout layer after the 3rd Convolutional block. Added 4th Convolutional block. |
| Binary Accuracy | 0.95 | 0.95 | 0.95 | 0.95 | 0.95 | 0.95 | 0.95 |
| AUC | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.98 | 0.99 |
| Optimal Threshold | 0.5 | 0.56 | 0.45 | 0.62 | 0.49 | 0.47 | 0.47 |
| Sensitivity | 0.96 | 0.96 | 0.97 | 0.97 | 0.97 | 0.97 | 0.96 |
| Specificity | 0.93 | 0.94 | 0.93 | 0.92 | 0.93 | 0.93 | 0.95 |
| Recall | 0.96 | 0.97 | 0.96 | 0.98 | 0.97 | 0.96 | 0.96 |
| Precision | 0.93 | 0.93 | 0.93 | 0.92 | 0.93 | 0.93 | 0.95 |
| Sensitivity at Specificity 0.95 | 0.91 | 0.93 | 0.93 | 0.90 | 0.92 | 0.91 | 0.95 |
| Specificity at Sensitivity 0.95 | 0.93 | 0.94 | 0.94 | 0.93 | 0.94 | 0.94 | 0.95 |

**Notes**: Sensitivity and Specificity are evaluated at the Optimal Threshold (knee of ROC curve); Recall and Precision are evaluated during the model fitting. AUC is Area Under the Curve
The best performing model has a ROC curve with a shape that allows the sensitivity to increase with the least amount of increase in false positives. The last two metrics are indicators of that tradeoff.

# Transfer Learning

Best Result

| Model | Xception | Xception V1 | EfficientNetB7 |
|---|---|---|---|
| Description | Xception feature extraction with 2 Dense classification layers. | Added a Dropout layer. Reduced Dense layer nodes. Included learning rate schedule and early stopping callbacks. | Includes a Dense layer with 256 nodes followed by a Dense layer with 1 output and sigmoid activation. Includes early stopping callback. |
| Binary Accuracy | 0.93 | 0.93 | 0.96 |
| AUC | 0.98 | 0.98 | 0.99 |
| Optimal Threshold | 0.29 | 0.45 | 0.61 |
| Sensitivity | 0.95 | 0.96 | 0.95 |
| Specificity | 0.93 | 0.92 | 0.96 |
| Recall | 0.91 | 0.94 | 0.96 |
| Precision | 0.95 | 0.93 | 0.95 |
| Sensitivity at Specificity 0.95 | 0.93 | 0.91 | 0.96 |
| Specificity at Sensitivity 0.95 | 0.93 | 0.92 | 0.96 |

**Notes**: Sensitivity and Specificity are evaluated at the Optimal Threshold (knee of ROC curve); Recall and Precision are evaluated during the model fitting. AUC is Area Under the Curve
The best performing model has a ROC curve with a shape that allows the sensitivity to increase with the least amount of increase in false positives. The last two metrics are indicators of that tradeoff.

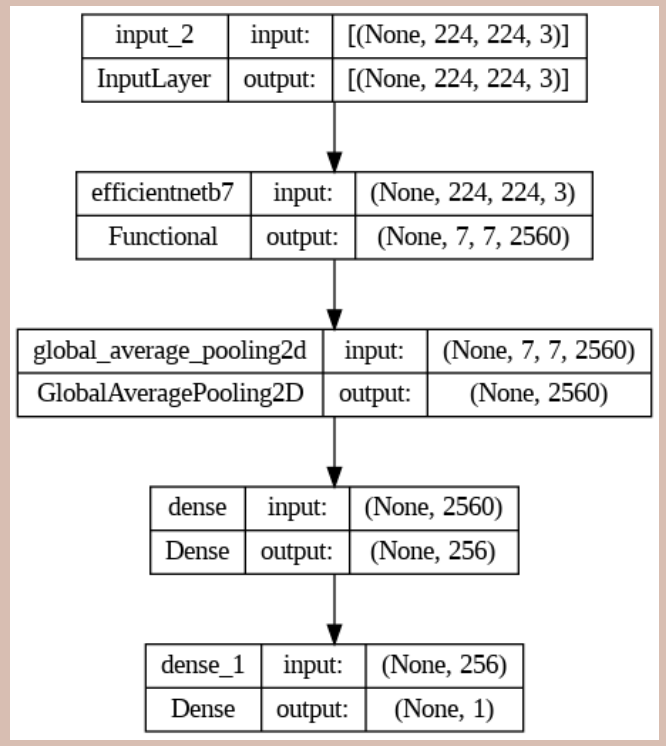
# Final Model Selection

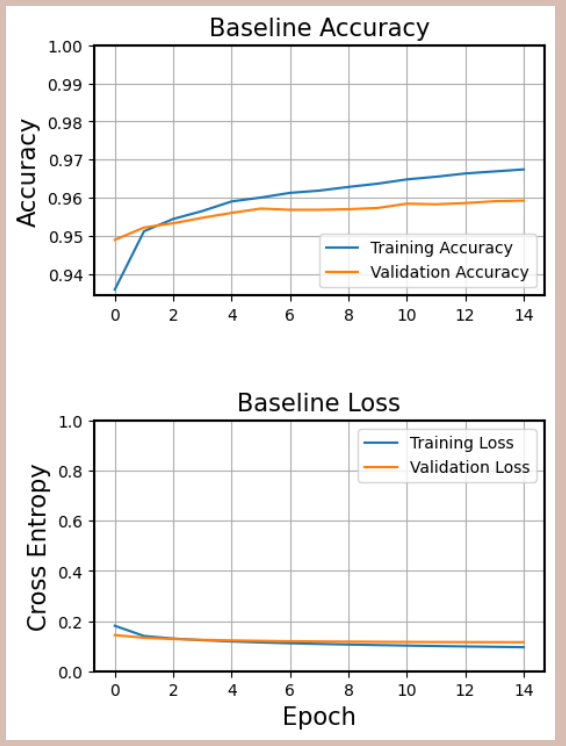
*Exploring Transfer Learning models*

- The EfficientNetB7 based model has the best performance. The architecture is shown here.
- This was trained with an early stopping callback set to monitor the validation loss. The training performance is shown here.

Pretrained Feature Extraction Layers

Classification Layers



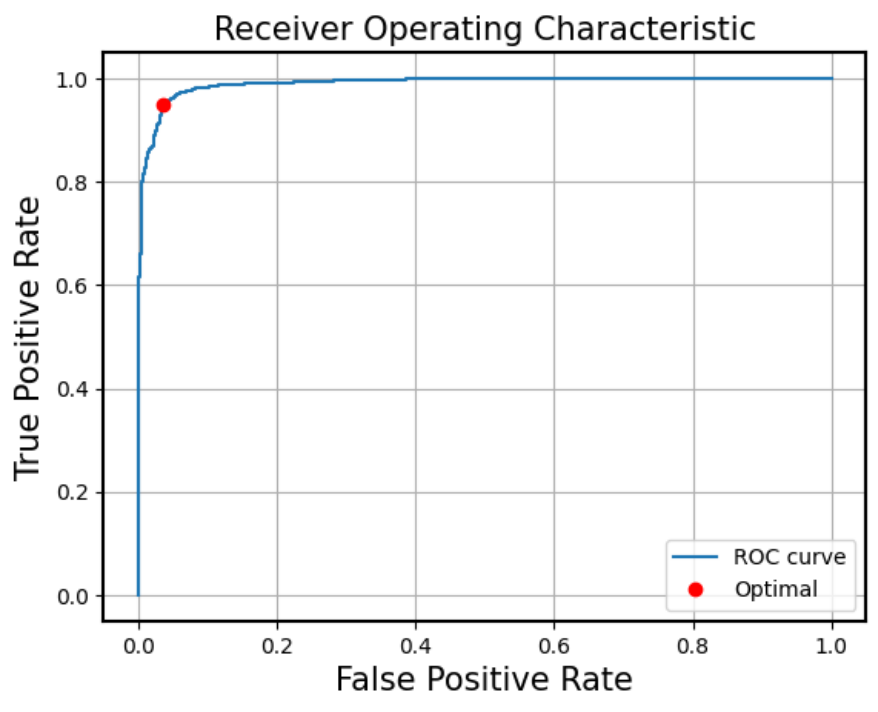



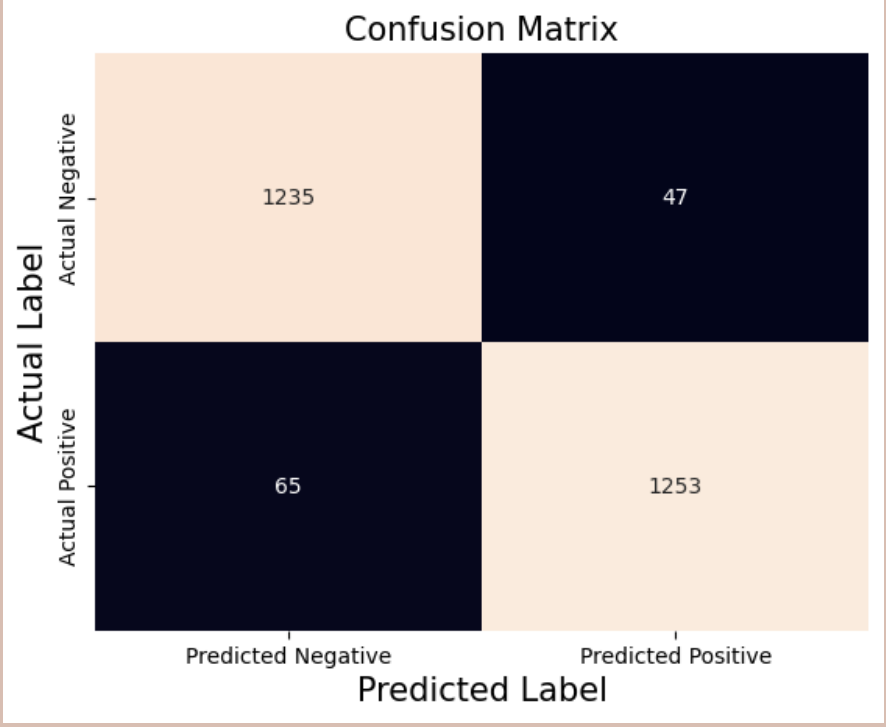

Several variations on Transfer Learning models and architectures were explored. The resulting models with the best performance were saved and the metrics tabulated. The EfficientNetB7 based model is the winner.

# Model Selection

## *Threshold selection*

The ROC curve is shown here with the optimal threshold / operating point. We can adjust the threshold to improve recall while maintaining a high specificity.



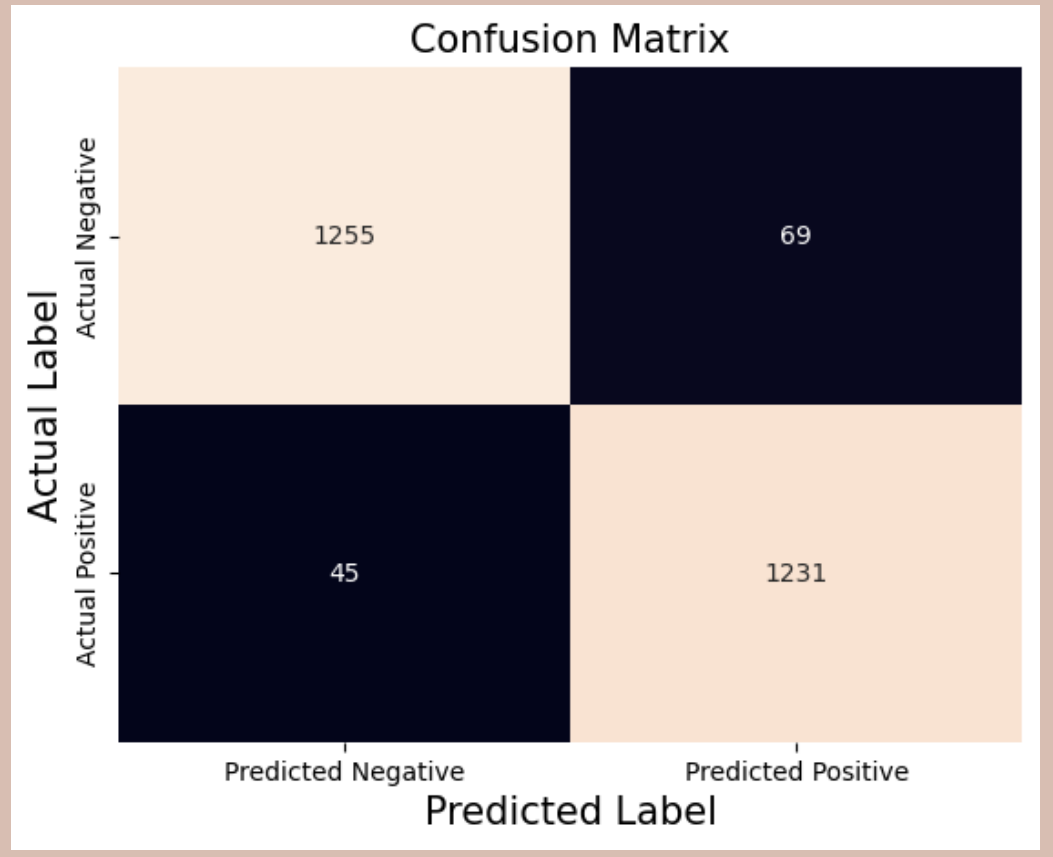




**Optimal Threshold = 0.61**
AUC: 0.99
True positives: 1235
False positives: 47
True negatives: 1253
False negatives: 65
Accuracy: 0.96
Sensitivity (TPR): 0.95
Specificity (TNR): 0.96
Precision: 0.96
F1 score: 0.96

# Model Selection

## *Threshold adjustment*

We can adjust the threshold to improve recall while maintaining a high specificity. This gives us the final selected operating threshold.




**Operating Threshold = 0.40**
AUC: 0.99
True positives: 1255
False positives: 69
True negatives: 1231
False negatives: 45
Accuracy: 0.96
Sensitivity (TPR): 0.97
Specificity (TNR): 0.95
Precision: 0.95
F1 score: 0.96

# Implications

*So what? How can this model be used?*

- We have shown a proof of concept. We have developed an efficient computer vision model to detect malaria and classify red blood cell images as parasitized or uninfected.
- The model could be deployed as part of a modified light microscopy system to automate or aid the detection and classification of the Plasmodium parasites that infect malaria patients' red blood cells.
- Automation can deployed in edge or IoT and cloud based implementations if internet access is available.
- Such an automation aided system would be scalable and part of regional laboratory services network and malaria case management and prevention programs.
- Automation can help improve accuracy and reduce variability of human readers. This could have a positive impact in endemic world regions where there is a lack of a skilled workforce or inadequate infrastructure.

# Some Insights and Next Steps

*Insights and Recommendations*

- We have seen some improvement in CNN performance can be made with image preprocessing methods including contrast enhancements and gamma corrections. These and other techniques could be evaluated with the transfer learning architecture developed here.
- We recommend developing concepts for an architecture that could be used to deploy an automated computer vision model, such as the one developed here, as part system that could augment or replace the need for microscopy laboratories for malaria surveillance and diagnosis.